Empirical results for Non-Parametric Maximum Likelihood Estimation (NPMLE)

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1 Problem Statement

$x_1, \ldots, x_N$ are drawn from some one-dimensional Gaussian Mixture distribution with unknown mean $\pi \sim \sum_{i=1}^{m} \alpha_i \mathcal{N}(\theta_i, 1)$, where $\sum_{i=1}^{m} \alpha_i = 1$, N is the number of samples, and m is the number of target mixtures. The Maximum Likelihood Estimation problem is to find

$$\hat{\pi}_{\text{NPMLE}} = \text{argmax}_{\pi} \sum \log L_{\pi}(x_i),$$

where $L_{\pi}(x_i)$ is the likelihood function of observing x_i in the distribution π.

For this report, the target distribution is mixture of Gaussian distributions with a finite but unknown support and variance one. So the problem is to find all means $\{\theta_i\}_m$. We are particularly interested in the self-regularizing effect in Non-parametric Maximum Likelihood Estimation (NPMLE), namely if the NPMLE will overfit the sample with significantly more supports.

2 Method

We follow the idea of vertex direction method (VDM) (cf. [2], [1], [3]). Instead of searching through all the distributions, we also use mixtures of Gaussians with finite support and variance one. On a high level, we find a new best direction in each iteration for step one. And for step two, we tune the weights of the mixtures. It is proved in the previous references that this is a consistent estimator. We will then describe more details about the algorithm.

Define the likelihood vector to be $\mathbf{f}_\theta(x) = (f_\theta(x_1), \ldots, f_\theta(x_N))$, where $f_\theta(x_i) = \mathcal{N}(x_i; \theta, 1)$ is the Gaussian likelihood center at θ_i. Similarly, the likelihood vector for the mixture is defined as $\mathbf{f}_\pi = \sum_{i=1}^{K} \alpha_i \mathbf{f}_{\theta_i}$. Then the log likelihood is $\phi(\mathbf{f}) = \sum_{i=1}^{N} \log f_i$.

It is shown [2] that the derivative of log likelihood with respect to a new direction θ is

$$D(\theta;\pi) = \sum_{i=1}^{N} \left(\frac{f_\theta(x_i)}{f_\pi(x_i)} - 1 \right) = \sum_{i=1}^{N} \omega_i f_\theta(x_i) - N, \tag{1}$$

where ω_i is some weights.

The the first step is the optimization problem

$$\hat{\theta} = \text{argmax}_\theta \, D(\theta;\pi) \tag{2}$$

Unfortunately, this optimization is not convex. So as a heuristic to conquer this, we experiment several points on equally spaced points and pick the best one to feed into the optimizer. This trick is only applicable when the dimension is low because the time complexity grows exponentially with the increase of dimension.

After finding the locally best theta, we need to adjust the weights α_i of the mixtures. Here we diverge slightly from the traditional VDM. In VDM, it searches a step size ϵ to maximize the log likelihood of the new estimator $(1-\epsilon)\pi + \epsilon\hat{\theta}$. In our case, we optimize the problem

$$\hat{\alpha} = \text{argmax}_\alpha \sum_{i=1}^{N} \log(\sum_{j=1}^{K} \alpha_j f_{\pi_j}(x_i) + \alpha_{K+1} f_{\hat{\theta}}(x_i)), \tag{3}$$

where π_j is the mean of previous estimator. The second step is a concave maximization.

Then, we iterate step 1 (Eq. 2) and step 2 (Eq. 3) until the increment of log likelihood is smaller then some threshold. The whole algorithm is described in Algorithm 1.

Algorithm 1 Algorithm (VDM like)

Require: θ_0 (Initialization)
  $\alpha_0 = (1)$
  $t = 0$
  while Increment in log likelihood is not small **do**
    $\theta_0 = \text{argmax}_{\theta \in \text{linspace}(-5,5,100)} \, D(\theta;\pi_t)$
    $\hat{\theta} = \text{argmax}_\theta \, D(\theta;\pi_t)$ (Use optimizer to find the maximum by the init θ_0)
    $\theta_{t+1} = \theta_t + (\hat{\theta})$ (Update the new mean)
    $\alpha_{t+1} = \text{argmax}_\alpha \, \phi(\mathbf{f}_{\theta_{t+1},\alpha})$
    $t = t + 1$
    Evaluate the log likelihood.
  end while

3 Experiments

Since we are interested in the self-regularizing effect, one important variable is the number of supports. Another metric we care is the 1-Wasserstein distance

because it considers the Euclidean distance in the distribution, not only the information, which may be misleading in our context.

3.1 One mixture

The target distribution is just the standard Gaussian $\mathcal{N}(0, 1)$. The initialized θ_0 is 1, so it doesn't start at the target theta. The number of Monte Carlo simulation is 1000.

The results are shown in Figure. 1, 2, 3. See more details in the captions.

3.2 Two mixtures

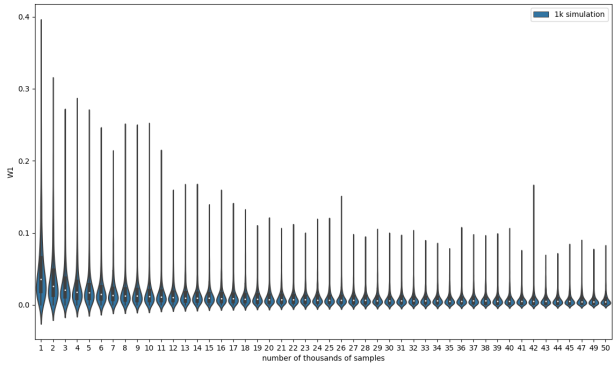
The target distribution have means $1, -1$ with 0.5 weight for each mean. The initialized θ_0 is drawn from standard Gaussian. The results are shown in Figure. 4, 5, 6.

References

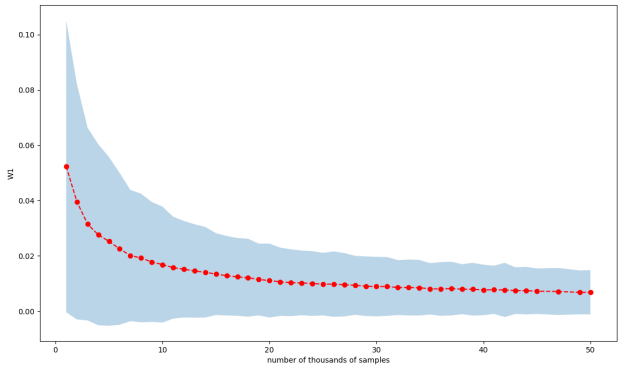
[1] Dankmar Bohning. "Convergence of Simar's algorithm for finding the maximum likelihood estimate of a compound Poisson process". In: *The Annals of Statistics* 10.3 (1982), pp. 1006–1008.

[2] Bruce G Lindsay. "The geometry of mixture likelihoods: a general theory". In: *The annals of statistics* (1983), pp. 86–94.

[3] Chien-Fu Wu. "Some algorithmic aspects of the theory of optimal designs". In: *The Annals of Statistics* (1978), pp. 1286–1301.

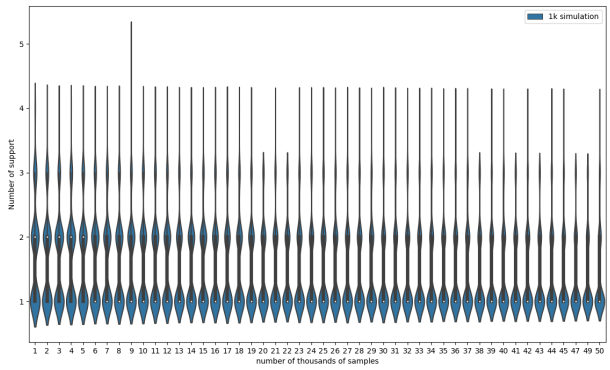


(a) Violin plot

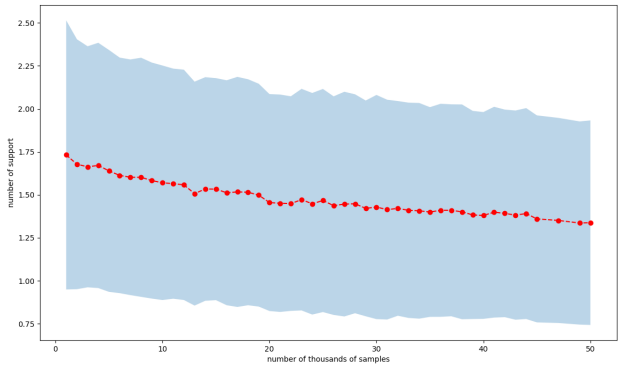


(b) Line plot

Figure 1: 1-Wasserstein distance VS number of thousands of samples for one mixture estimation.

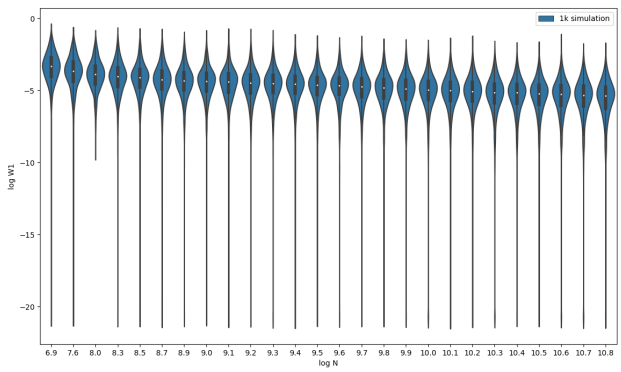


(a) Violin plot. The number of support is discrete so there are lumps of data. With the increase of samples, the lumps transfer mass to lower lumps.

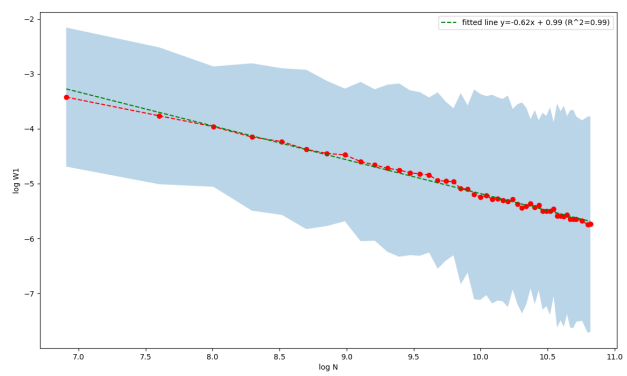


(b) Line plot. The average number of support decreases constantly.

Figure 2: number of supports VS number of thousands of samples for one mixture estimation.

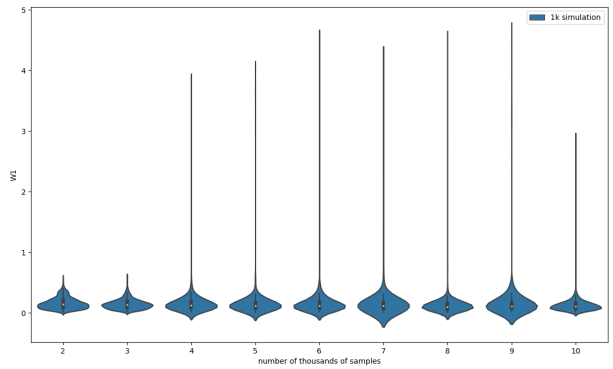


(a) Violin plot

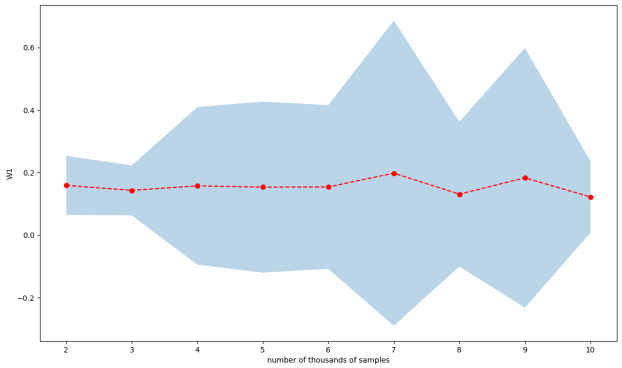


(b) Line plot. The green line is the fitted line and the red line is the line plot of the actual data and they fit pretty well. Interestingly, the slope of decrease is smaller than 0.5, the theoratical lower bound.

Figure 3: number of supports VS number of thousands of samples for one mixture estimation.

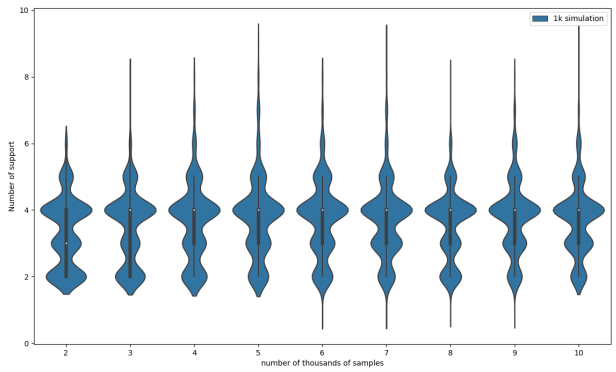


(a) Violin plot



(b) Line plot

Figure 4: 1-Wasserstein distance VS number of thousands of samples for two mixture estimation.



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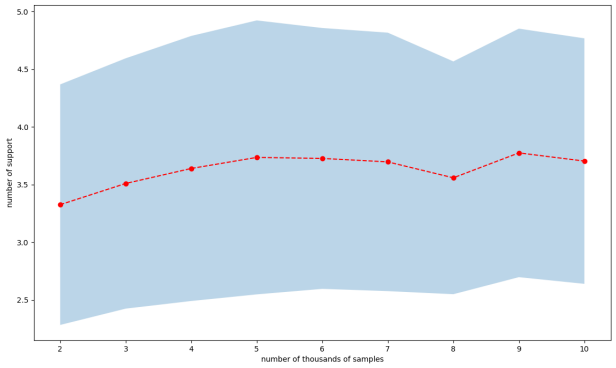
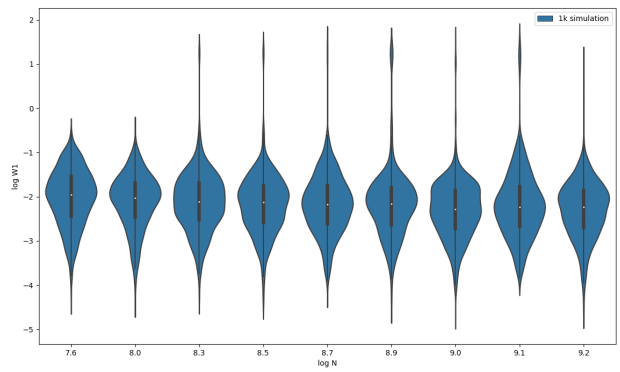
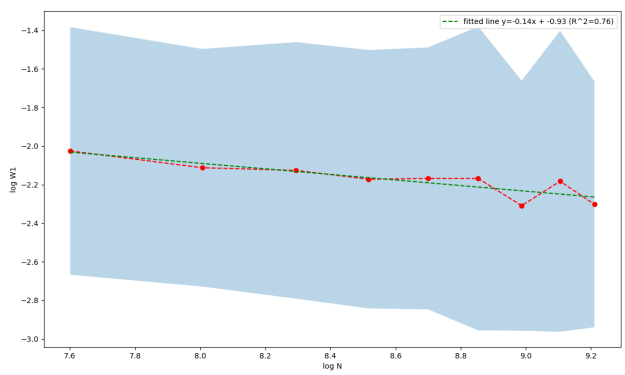


(b) Line plot. The average number of support decreases constantly.

Figure 5: number of supports VS number of thousands of samples for two mixture estimation.



(a) Violin plot



(b) Line plot. The green line is the fitted line and the red line is the line plot of the actual data and they fit pretty well. The slope of decrease is very slow.

Figure 6: number of supports VS number of thousands of samples for two mixture estimation.